

PatchPad Editor v3 - Evaluation Report

Harbor Web Development Task | Oracle, target-model and comparison-run evidence

Harbor task: turing/patchpad-editor-v3 | version 1.0.0 | prepared 13 September 2026

Scope: recorded platform outcomes. GPT-5.4 mini is the score target; Gemini has no score restriction. The user authorized final packaging with the reported evidence limitations retained. No scores, task requirements or verifier weights were changed.

Headline

Oracle scored 1.0000 and passed all 41 criteria. GPT-5.4 mini (high), the target model, scored 0.5605 within the 0.10-0.70 band. Gemini completed at 0.7622 with no score restriction. Haiku completed at 0.2233 with its Functional browser gate failed. The no-op control scored 0.0000. All four evaluated apps have graded=1, no_op=0 and no trial exception.

Title, runtime and evaluated features

PatchPad is a custom incident-report editor with exact long-document editing, find/replace, clipboard operations, multiple carets, atomic history and conflict-safe SQLite revisions. The golden app uses a custom DOM surface and Node built-in SQLite, started by npm start; its manifest declares /app/data/patchpad.db. The agent may choose its own custom implementation within the five supplied specifications.

Area	Coverage
Custom editor and seed	Custom DOM/canvas/SVG document surface; supplied Northwind API Incident Report; 1,226 logical lines, original metadata and revision 1.
Editing and selection	Logical line/cursor positions, word navigation, grapheme-safe Unicode, mouse word/line/range selection, offscreen selection and multiple carets.
Clipboard and history	Exact multiline/tabbed copy-cut-paste, typing groups, atomic selection replacement, block indentation, Replace All, Undo/Redo and redo-branch invalidation.
Saved revisions	Explicit Save, no-op save invariants, reload/fresh-client persistence, revision preview and unsaved restore with Undo.
Server safety	SQLite durability, restart seed idempotence, chained two-tab stale-save conflicts and malformed/contradictory save rejection without mutation.
Interface	Discoverable controls, focus and readable feedback; five visual axes at 1280 by 800. No mobile or sign-in requirement.

Five batched browser dimensions use Codex / gpt-5.6-luna, max reasoning, temperature 0 and Playwright MCP. There are 41 criteria: 2 Render, 2 Constraints, 29 Functional, 3 Polish and 5 Visual. Visual reviews only the 1280 by 800 desktop layout. Agent and verifier networking are public. Authentication and mobile layouts are not requirements.

Overall scores

Run	Overall	Render	Constr.	Func.	Polish	Visual	Recorded status
Oracle	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	29/29 Functional
GPT-5.4 mini (high)	0.5605	1.0000	1.0000	0.4286	0.6667	0.8500	12/29 Functional
Gemini 3.7 Flash (high)	0.7622	1.0000	1.0000	0.6703	1.0000	0.8000	22/29 Functional

Run	Overall	Render	Constr.	Func.	Polish	Visual	Recorded status
Claude Haiku 4.5	0.2233	1.0000	1.0000	0.0000	0.6667	0.4500	Functional gate failed; 29 checks skipped
No-op control	0.0000	0	0	0	0	0	graded=0; no_op=1

Binary pass counts differ from weighted scores. Haiku Functional zeros are gate assignments, not independently executed checks. Visual raw ratings use the installed runner normalization: 4 becomes 0.75 and 5 becomes 1.00. The no-op zero is an ungraded floor control, not a model failure.

Reward arithmetic

If Render <= 0 or Constraints <= 0: Overall = 0. Otherwise: Overall = round(0.6 x Functional + 0.2 x Polish + 0.2 x Visual, 4).

Run	Functional weight earned	Final calculation
Oracle	22.75 / 22.75 = 1.0000	0.6 x 1.0000 + 0.2 x 1.0000 + 0.2 x 1.0000 = 1.0000
GPT-5.4 mini (high)	9.75 / 22.75 = 0.4286	0.6 x 0.4286 + 0.2 x 0.6667 + 0.2 x 0.8500 = 0.5605
Gemini 3.7 Flash (high)	15.25 / 22.75 = 0.6703	0.6 x 0.6703 + 0.2 x 1.0000 + 0.2 x 0.8000 = 0.7622
Claude Haiku 4.5	0 / 22.75 = 0.0000	0.6 x 0.0000 + 0.2 x 0.6667 + 0.2 x 0.4500 = 0.2233

Dimension scores and budgets

Dimension	Criteria	Budget	Role	Oracle	GPT-mini	Gemini	Haiku
Render	2	600 s	Gate	1.0000	1.0000	1.0000	1.0000
Constraints	2	600 s	Gate	1.0000	1.0000	1.0000	1.0000
Functional	29	9000 s	60%	1.0000	0.4286	0.6703	0.0000
Polish	3	900 s	20%	1.0000	0.6667	1.0000	0.6667
Visual	5	900 s	20%	1.0000	0.8500	0.8000	0.4500

All 41 verifiers - recorded results

These are exported verdicts, not substituted local grades. Each evaluated application has an outcome for every criterion; a gate-assigned Fail does not imply the behavior was individually exercised.

Render

Criterion (underscores shown as spaces)	Weight	Oracle	GPT-mini	Gemini	Haiku
page load and refresh	1.0	Pass	Pass	Pass	Pass
basic control interaction	1.0	Pass	Pass	Pass	Pass

Constraints

Criterion (underscores shown as spaces)	Weight	Oracle	GPT-mini	Gemini	Haiku
local application entry	1.0	Pass	Pass	Pass	Pass
self contained entry and reload	1.0	Pass	Pass	Pass	Pass

Functional

Criterion (underscores shown as spaces)	Weight	Oracle	GPT-mini	Gemini	Haiku
seed document integrity	0.25	Pass	Fail	Pass	Fail
custom editor surface real input	0.125	Pass	Pass	Pass	Fail
unsaved edit discard on reload	0.125	Pass	Pass	Pass	Fail
noop save revision invariant	0.5	Pass	Pass	Pass	Fail
edit save reload and fresh client	0.25	Pass	Fail	Pass	Fail
keyboard navigation exact coordinates	1.0	Pass	Fail	Fail	Fail
word navigation and selection shortcuts	0.5	Pass	Fail	Pass	Fail
backspace delete exact line join	0.5	Pass	Fail	Pass	Fail
tab indentation and reversal	0.5	Pass	Fail	Pass	Fail
unicode grapheme backspace delete	0.5	Pass	Fail	Pass	Fail
unicode grapheme navigation selection	0.5	Pass	Fail	Pass	Fail
selection real mouse word line range keyboard	1.5	Pass	Fail	Fail	Fail
selection autoscroll exact offscreen range	1.0	Pass	Fail	Fail	Fail
clipboard external multiline internal exact	1.0	Pass	Pass	Pass	Fail
undo separate locations and redo invalidation	1.0	Pass	Fail	Pass	Fail
undo paste cut atomic	1.0	Pass	Fail	Pass	Fail

Criterion (underscores shown as spaces)	Weight	Oracle	GPT-mini	Gemini	Haiku
undo typed selection replacement atomic	1.5	Pass	Pass	Pass	Fail
find replace exact counts and offsets	0.5	Pass	Fail	Fail	Fail
keyboard find focus and cycle	0.5	Pass	Pass	Pass	Fail
long document three region round trip	0.5	Pass	Pass	Pass	Fail
revision history preview restore undo exact	1.5	Pass	Pass	Fail	Fail
two tab chained stale save conflicts	1.5	Pass	Pass	Pass	Fail
multi caret full typing single undo	1.5	Pass	Fail	Fail	Fail
multi caret backspace delete sibling	0.5	Pass	Fail	Fail	Fail
block indentation atomic history	1.0	Pass	Fail	Pass	Fail
replace all atomic history and branch	1.0	Pass	Fail	Pass	Fail
manifest documents runtime routes	0.5	Pass	Pass	Pass	Fail
restart seed idempotence and saved history	1.0	Pass	Pass	Pass	Fail
direct api save rejection nonmutation matrix	1.0	Pass	Pass	Pass	Fail

Polish

Criterion (underscores shown as spaces)	Weight	Oracle	GPT-mini	Gemini	Haiku
labelled primary controls	1.0	Pass	Pass	Pass	Pass
keyboard focus and editor entry	1.0	Pass	Fail	Pass	Fail
history and feedback readability	1.0	Pass	Pass	Pass	Pass

Visual

Criterion (underscores shown as spaces)	Weight	Oracle	GPT-mini	Gemini	Haiku
visual typography	1.0	5/5	4/5	4/5	3/5
visual color and contrast	1.0	5/5	5/5	4/5	4/5
visual spacing and layout	1.0	5/5	5/5	5/5	2/5
visual hierarchy and scanability	1.0	5/5	4/5	4/5	2/5
visual overall craft	1.0	5/5	4/5	4/5	3/5

Oracle coverage

All 36 binary criteria passed and all five Visual criteria scored raw 5/5. Every Functional observation is present, including block indentation, Replace All atomic history, stale saves, rejected API probes and two managed restarts. This is a recorded full Oracle pass; preparation did not run a new paid judge.

GPT-5.4 mini - interpretation and evidence gaps

The official 0.5605 is inside the target band. Some failed checks never reached a valid action or complete comparison. This limits attribution; it does not prove the app would pass and does not justify changing the official score.

Criterion	Unresolved evidence
edit save reload and fresh client	The marker saved and reloaded, but the full pre-save draft comparison was not established.
backspace delete exact line join	Two attempts did not establish the required three adjacent lines, so the boundary edits were unverified.
multi caret full typing single undo	The judge did not establish three visible carets, so typing/Undo/Redo was unverified.
multi caret backspace delete sibling	The judge did not establish the second caret required for deletion.
replace all atomic history and branch	Replace All and single Undo/Redo worked, but the branch insertion and Redo invalidation checkpoint were not established.

The judge separately reported unpadded ALPHA markers, Shift+Tab joining the following line, and selected-block indentation altering the outside line. Missing seed markers also blocked the exact offscreen and separate-location setups; those observations share a seed defect and should not be presented as independent root bugs. No focused GPT browser regrade was performed during preparation.

Failed verifiers - exact exported judge explanations

GPT-5.4 mini (high)

Functional - recorded deductions

Criterion / recorded value	Exported judge explanation
seed document integrity Fail	Revision 1 had 1,226 lines, but required lines 24 and 623 used unpadded ALPHA-1 and ALPHA-600 instead of ALPHA-0001 and ALPHA-0600.
edit save reload and fresh client Fail	BASIC-SAVE-CHECK saved from revision 1 to 2 and loaded once in a fresh client, but complete pre-save draft comparison was not established from the virtualized DOM.
keyboard navigation exact coordinates Fail	Clicking visible first-line text placed the caret on displayed line 5, with gutter label 5, rather than the first logical line.
word navigation and selection shortcuts Fail	NORTH and WIND copied correctly, but modifier+Left after End stayed at column 11 instead of the required column 7.
backspace delete exact line join Fail	Two setup attempts failed to establish three adjacent JOIN-LEFT/JOIN-RIGHT/JOIN-TAIL lines, so the required boundary edits were not verified.
tab indentation and reversal Fail	Tab produced exact <code>`\tTimeline`</code> , but Shift+Tab joined the following line instead of restoring exact Timeline.
unicode grapheme backspace delete Fail	The exact ☺ and decomposed é selections copied correctly, but one Backspace and one Delete left both original lines unchanged.
unicode grapheme navigation selection Fail	The exact A☺éB sample was found, but the required sequence copied <code>`:`</code> , <code>`☺`</code> , and <code>`B`</code> instead of A, ☺, and é.

Criterion / recorded value	Exported judge explanation
selection real mouse word line range keyboard Fail	Real double/triple clicks and the drag selected the wrong rendered text/range; only the later keyboard replacement produced the required DONE line.
selection autoscroll exact offscreen range Fail	Required ALPHA-0010 and ALPHA-0013 markers were absent; the live document had ALPHA-10 and ALPHA-13, so setup was unavailable.
undo separate locations and redo invalidation Fail	The required ALPHA-0600 target was absent. On ALPHA-600 the full LOCATION-ONE/TWO/THREE Undo/Redo sequence worked, but the named setup could not be established.
undo paste cut atomic Fail	PASTE-A/B/C paste, cut, Undo, and Redo mostly worked, but the first Undo failed to restore the recorded empty final line.
find replace exact counts and offsets Fail	NEXT had 3 matches and cycled 19→20→18, but exact clipboard checkpoints were not resolved; ALPHA-00 reported 0 instead of 99.
multi caret full typing single undo Fail	Real modifier-click attempts did not establish exactly three visible carets, so the required MULTI typing, Undo, and Redo observations were unavailable.
multi caret backspace delete sibling Fail	The real normal click plus modifier-click did not establish the second required end-of-line caret, so neither multi-caret deletion was verified.
block indentation atomic history Fail	The first Tab indented BLOCK-ONE and BLOCK-TWO but joined BLOCK-THREE with BLOCK-OUTSIDE, changing the fourth line.
replace all atomic history and branch Fail	Replace All changed all three ATOMIC-OLD occurrences to ATOMIC-NEW and single Undo/Redo worked, but the required BRANCH-NEW insertion/setup and Redo invalidation were not established.

Polish - recorded deductions

Criterion / recorded value	Exported judge explanation
keyboard focus and editor entry Fail	Toolbar buttons, both inputs, and history controls were keyboard-reachable with visible focus outlines; Escape from the editor moved focus to Find without changing content. However, the focusable custom editor itself had no visible outline and no caret on fresh keyboard focus (computed outline was none), so not every focused control met the requirement.

Visual - recorded deductions

Criterion / recorded value	Exported judge explanation
visual typography 4/5	Crisp sans-serif UI text and a readable monospace editor font are consistently aligned across the header, toolbar, gutter, history, and preview. The report title is slightly understated relative to surrounding hierarchy.
visual hierarchy and scanability 4/5	Brand, report selection, controls, document, status, and revision history are easy to scan, with saved feedback clearly highlighted. The report title and primary Save action could have stronger visual emphasis.
visual overall craft 4/5	The custom editor, buttons, cards, gutter, revision history, and preview modal share a polished visual language with no stray default styling. Minor polish gaps are the understated report heading and slightly oversized indentation before document text.

Gemini 3.7 Flash (high)

Functional - recorded deductions

Criterion / recorded value	Exported judge explanation
keyboard navigation exact coordinates Fail	Keyboard coordinates and gutter labels passed, but final-line mouse placement resolved to visible line 42 instead of logical line 1,226.
selection real mouse word line range keyboard Fail	Word and line gestures passed, but the real drag selected "eckout requests " instead of exactly "Checkout requests".
selection autoscroll exact offscreen range Fail	Mouse selection scrolled but ended with a truncated ALPHA-0013 range; keyboard setup also failed the required initially-offscreen condition.
find replace exact counts and offsets Fail	NEXT count and 19→20→18 cycling passed, but replacement and Replace-All evidence was invalidated by focus/selection setup errors.
revision history preview restore undo exact Fail	History and revision-1 preview passed, but the required Restore dirty-state and one-Undo checkpoint was lost to a judge result-assembly error.
multi caret full typing single undo Fail	Three real Alt-click carets inserted MULTI correctly, but one Undo left MULT at all three carets instead of removing the full group.
multi caret backspace delete sibling Fail	Two-carets Backspace passed, but start placement was column 2, producing Cstomer/Ceckout instead of utomer/heckout on Delete.

Visual - recorded deductions

Criterion / recorded value	Exported judge explanation
visual typography 4/5	Readable monospace editor text and a clear sans-serif header/UI system. Section labels remain the same weight and size as body text, and the report selector truncates its selected text.
visual color and contrast 4/5	Cohesive navy/black palette with legible white text, blue actions, green saved states, and orange search highlights. History metadata, footer shortcuts, and disabled controls are noticeably muted.
visual hierarchy and scanability 4/5	Brand, report identity, primary actions, save feedback, history, and preview controls are easy to locate. Document section headings are visually understated, so scanning relies heavily on blank-line separation.
visual overall craft 4/5	Editor, Find/Replace, history, and preview share a polished dark IDE-like design language with integrated buttons, cards, status pills, and modal treatment. Minor polish gaps include plain document headings, subdued secondary text, and selector truncation.

Claude Haiku 4.5

Functional - recorded deductions

Criterion / recorded value	Exported judge explanation
seed document integrity Fail	Global browser gate failed: editor text/caret were offscreen (#editor x=1268, editor wrapper width=0), and a real editor click was intercepted. Seed response also had unpadding Log line 1/600.
custom editor surface real input Fail	Global browser gate failed; criterion not scored.

Criterion / recorded value	Exported judge explanation
unsaved edit discard on reload Fail	Global browser gate failed; criterion not scored.
noop save revision invariant Fail	Global browser gate failed; criterion not scored.
edit save reload and fresh client Fail	Global browser gate failed; criterion not scored.
keyboard navigation exact coordinates Fail	Global browser gate failed; criterion not scored.
word navigation and selection shortcuts Fail	Global browser gate failed; criterion not scored.
backspace delete exact line join Fail	Global browser gate failed; criterion not scored.
tab indentation and reversal Fail	Global browser gate failed; criterion not scored.
unicode grapheme backspace delete Fail	Global browser gate failed; criterion not scored.
unicode grapheme navigation selection Fail	Global browser gate failed; criterion not scored.
selection real mouse word line range keyboard Fail	Global browser gate failed; criterion not scored.
selection autoscroll exact offscreen range Fail	Global browser gate failed; criterion not scored.
clipboard external multiline internal exact Fail	Global browser gate failed; criterion not scored.
undo separate locations and redo invalidation Fail	Global browser gate failed; criterion not scored.
undo paste cut atomic Fail	Global browser gate failed; criterion not scored.
undo typed selection replacement atomic Fail	Global browser gate failed; criterion not scored.
find replace exact counts and offsets Fail	Global browser gate failed; criterion not scored.
keyboard find focus and cycle Fail	Global browser gate failed; criterion not scored.
long document three region round trip Fail	Global browser gate failed; criterion not scored.
revision history preview restore undo exact Fail	Global browser gate failed; criterion not scored.
two tab chained stale save conflicts Fail	Global browser gate failed; criterion not scored.
multi caret full typing single undo Fail	Global browser gate failed; criterion not scored.
multi caret backspace delete sibling Fail	Global browser gate failed; criterion not scored.
block indentation atomic history Fail	Global browser gate failed; criterion not scored.
replace all atomic history and branch Fail	Global browser gate failed; criterion not scored.
manifest documents runtime routes Fail	Global browser gate failed; criterion not scored.
restart seed idempotence and saved history Fail	Global browser gate failed; criterion not scored.
direct api save rejection nonmutation matrix Fail	Global browser gate failed; criterion not scored.

Polish - recorded deductions

Criterion / recorded value	Exported judge explanation
keyboard focus and editor entry Fail	Tab reaches toolbar controls and the search inputs/buttons have visible focus indicators. However, after keyboard entry into the editor, Escape leaves focus on the editor textarea instead of moving to the Find input.

Visual - recorded deductions

Criterion / recorded value	Exported judge explanation
visual typography 3/5	Header, toolbar, status, and preview text are readable with clear title/label sizing. The main editor’s line numbers form a dense horizontal wall while report text is not visibly readable, weakening consistency.
visual color and contrast 4/5	The light neutral palette with blue actions, green Saved status, and readable preview contrast is cohesive. The very light gutter and unreadable editor body reduce legibility on the primary surface.
visual spacing and layout 2/5	Header, toolbar, history tray, and centered preview are neatly aligned, but the primary editor is severely broken: line numbers wrap across the document area and the report text column is clipped or out of view.
visual hierarchy and scanability 2/5	Title, metadata, Saved badge, toolbar, and preview/history headings scan clearly. However, the main report cannot be scanned because the editor body is obscured by the repeated-number gutter.
visual overall craft 3/5	The blue/gray component language, borders, history panel, and revision preview are reasonably coherent and polished. The primary editor’s broken gutter and missing visible text make the overall interface feel unfinished.

Comparison-run interpretation

Gemini has no score restriction. It completed normally at 0.7622; its higher score is retained as comparison evidence. The judge explicitly lost the revision Restore/Undo checkpoint to a result-assembly error, and other deductions cite focus/selection setup problems. These do not establish that every failed criterion is an app defect.

A focused offline diagnostic of unchanged Gemini source reproduced multi-caret Undo leaving MULT at all three positions after typing MULTI. Its preview of revision 1, unsaved Restore and single Undo passed complete content comparisons without changing saved content or history. Four local attempts were retained in the workspace review, including focus/coordinate setup failures; the final diagnostic tested typing groups at observed interior positions, not proof of correct start-of-line placement. It used a cached verifier image and is not an official regrade.

Haiku Functional failed its global browser gate: the judge described zero editor width, offscreen text/caret and an intercepted click. It then assigned zero to all 29 criteria without individually testing their behavior. Render/Constraints credited Find interactions and Visual awarded partial credit despite reporting clipped report text. The disagreement remains documented; the final 0.2233 is not silently changed.

Run index and evidence locations

Oracle | run-08e5c411-7f8e-40a4-afac-47689b9d832a | patchpad-editor-v3__KWEbkhe

Archive: patchpad-editor-v3-oracle-job-directory.zip. Inside its matching wrapper, patchpad-editor-v3__KWEbkhe/verifier/reward.json gives the authoritative score; reward-details.json gives all criterion reasons; prompt-provenance.json gives verifier hashes. Trial result.json records completion and task checksum. Builder trajectories are in agent/trajectory.json for model runs.

GPT-5.4 mini (high) | run-19fa3352-989a-4306-bc14-fc57cca17d9b | patchpad-editor-v3__XDzZG5e

Archive: patchpad-editor-v3-gpt-5.4-mini-high-job-directory.zip. Inside its matching wrapper, patchpad-editor-v3__XDzZG5e/verifier/reward.json gives the authoritative score; reward-details.json gives all criterion reasons; prompt-provenance.json gives verifier hashes. Trial result.json records completion and task checksum. Builder trajectories are in agent/trajectory.json for model runs.

Gemini 3.7 Flash (high) | run-94c32bc4-7dcc-4574-b511-b11e07b71564 | patchpad-editor-v3__yeYzfVZ

Archive: patchpad-editor-v3-gemini-3.7-flash-job-directory.zip. Inside its matching wrapper, patchpad-editor-v3__yeYzfVZ/verifier/reward.json gives the authoritative score; reward-details.json gives all criterion reasons; prompt-provenance.json gives verifier hashes. Trial result.json records completion and task checksum. Builder trajectories are in agent/trajectory.json for model runs.

Claude Haiku 4.5 | run-918618b3-4e1d-47da-988e-7e306001ec0a | patchpad-editor-v3__iP3FvE3

Archive: patchpad-editor-v3-claude-haiku-4.5-job-directory.zip. Inside its matching wrapper, patchpad-editor-v3__iP3FvE3/verifier/reward.json gives the authoritative score; reward-details.json gives all criterion reasons; prompt-provenance.json gives verifier hashes. Trial result.json records completion and task checksum. Builder trajectories are in agent/trajectory.json for model runs.

The Oracle job archive also includes patchpad-editor-v3__eNxTkub, the no-op trial, plus the original job-level config, lock, result and logs. There is one supplied Oracle, one no-op and one trial for each comparison/target model; no trial was dropped to select a more favorable score.

Delivery and provenance

The final folder follows the supplied seven-file delivery layout: unchanged task ZIP, Oracle/no-op job-directory ZIP, GPT-mini job-directory ZIP, Gemini job-directory ZIP, Haiku job-directory ZIP, this case study and the companion evaluation report. Job exports preserve recorded scores, criterion reasons, source and trajectories. Runtime databases and caches are excluded; sanitization is packaging-only.

All five trials share task checksum 132a2f0fb7ae8e5a9a02f154fe7d5de1d2c232c7aaf45e3e3f0c5d057dc9f659. All logged verifier hashes match the packaged source. All 32 task archive files match current source; the task ZIP checksum is a3511a525675554aaa1ee89b5e77fb69ad4f7a94a1fc68b7d137bc39053439a6.

No deliberate cheating was found in the reviewed source and builder trajectories. The scan found no matching credential tokens or grader-directed tampering. This is a scoped assessment, not an absolute guarantee. Native judge action traces and a platform QC report were not present in the supplied exports. Packaging does not claim either has been produced.

The user authorized this final package after review of GPT evidence gaps. Scores and requirements remain unchanged; Gemini is not subject to the target score band. The evaluation report contains every criterion outcome and the recorded failure explanations, with an index to the included jobs.